th National High School Physics Finals: Solutions

I. (points) Fill in the blanks

points)

points)

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points)

II. (points)

Let be the block mass. Its acceleration down the incline is

The speed of the block at the bottom of the incline is

In the belt frame, the initial speed of the block entering the belt is

The angle between the motion and the belt edge satisfies

The block decelerates on the belt with acceleration magnitude

The distance traveled relative to the belt before coming to rest is

The least friction coefficient preventing escape over the belt edge satisfies